EDGE AI & THE RASPBERRY PI

1. By the end of today I can...

- [ ] explain Edge AI in two sentences
- [ ] name the main parts of the Pi 4
- [ ] write the OS onto a microSD card with my own settings
- [ ] power the Pi, read its 2 LEDs, wait out first boot
- [ ] log in over **SSH** from Windows and shut down safely

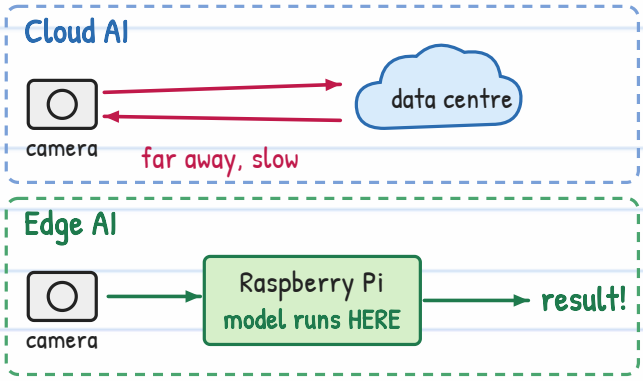
My 2 hours

A big picture	20 min
B meet the board	20 min
C prepare microSD	45 min
D first boot	15 min
E first SSH login	20 min

my notes

2. What is Edge AI?

- Cloud AI: my data travels to a **data centre** and the answer travels back.
- Edge AI = the model runs ON the device next to the data (camera, robot, sensor box, a Pi).
- Book: PDF pp. 20–21 and 25–27 (printed page + 1).



my notes

3. Five reasons it matters

Advantage	What it means in practice
Low latency	answer in milliseconds, no round trip
Privacy	raw images / audio stay on the device
Network independence	works in a tunnel, a field, a factory
Less bandwidth	send a result, not a video stream
Energy efficiency	small optimised models, small power

my notes

AI FAMILIES & MEET THE BOARD

4. Two families of AI in this course

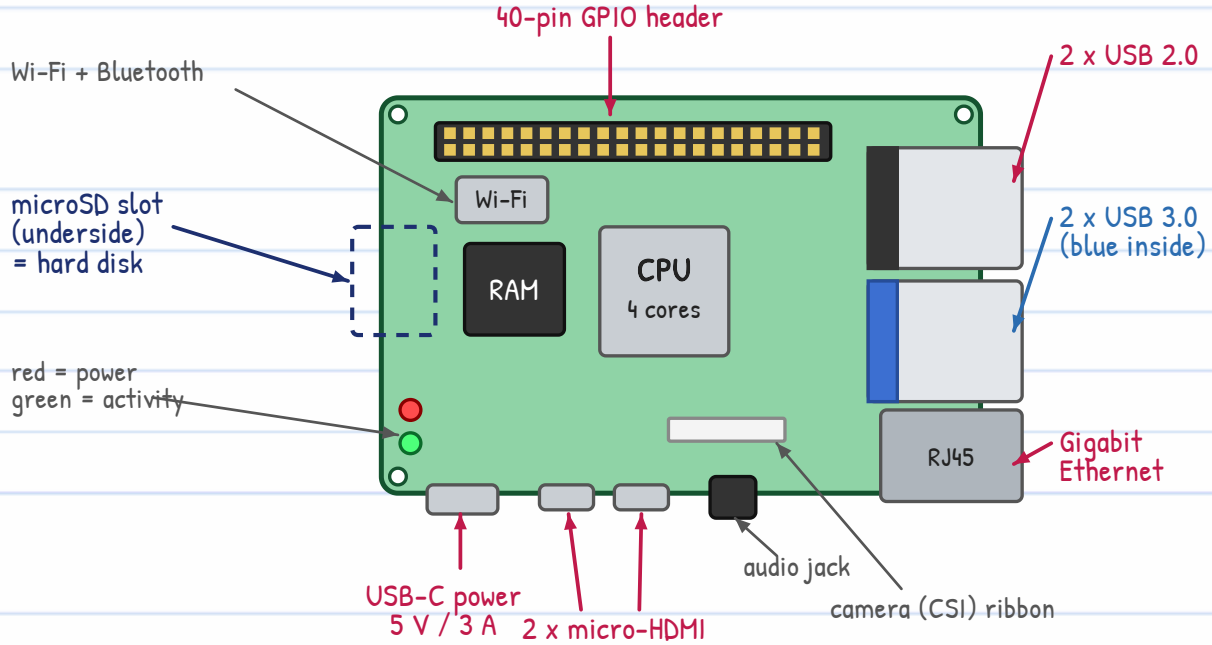
Fixed-function (reactive)
input -> fixed kind of answer: image classification, object detection, counting with YOLO.
Phase 2, days 14-26

Generative (proactive)
creates new content: small language models, vision-language models, agents.
Phase 3, days 27-38

My board = Pi 4, 4 GB. Vision runs fine. Language models: only the small ones, slower than the book's Pi 5. RAM is the limit.

my notes

5. Anatomy of the Raspberry Pi 4



(schematic - port positions simplified)

Part	What it does	Why I care
CPU, 4 cores	runs the OS + my code	does the neural-network maths
RAM 4 GB	working memory	models must fit in it
microSD	the "hard disk"	always shut down first
GPIO, 40 pins	sensors, LEDs, buttons	physical computing (Phase 3)
CSI / USB	camera ports	vision projects (Day 4)

Book PDF pp. 29-33. The book's picture is the Pi 5 (power button + PCIe; mine has neither).

my notes

6. My turn

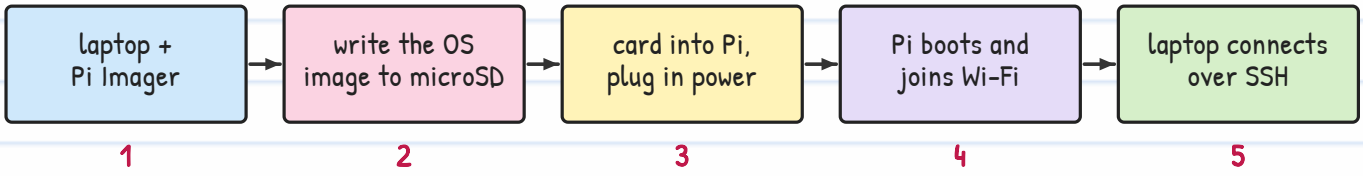
- [ ] find every part on my real board, take a top-down photo, label it
- [ ] hold the board by its edges - no metal table, no wet fingers

my notes

PREPARE THE MICROSD CARD

7. The idea

- The Pi has **no hard disk**. It boots from the **microSD**.
- "Installing the OS" = writing an image onto the card with **Raspberry Pi Imager**.
- I also bake in my **settings** (name, user, Wi-Fi, SSH): no monitor or keyboard needed = **headless**.



my notes

8. Which operating system?

Choice in Imager	What it is	Mine?
Raspberry Pi OS (64-bit)	full system **with desktop**; 64-bit needed by AI libraries	YES
Raspberry Pi OS Lite (64-bit)	no desktop, uses less RAM (book uses it on the Pi Zero)	not today
32-bit versions	older, less compatible	avoid

The book says "Debian Bookworm". A newer Imager may offer a newer Debian. Steps are the same, but the **Python version** may differ and TFLite is picky - I record it today and check on Day 4-5.

my notes

9. Before I start

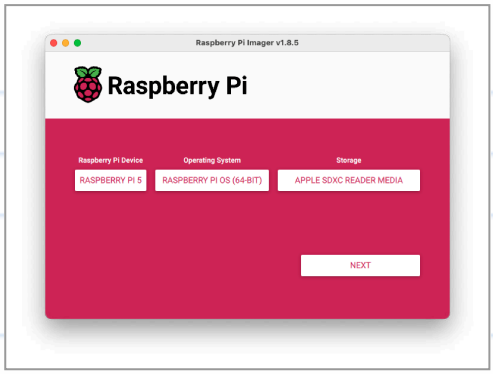
- [ ] Raspberry Pi Imager installed (raspberrypi.com/software)
- [ ] Wi-Fi name + password ready
- [ ] card backed up - Imager **erases** it
- [ ] other USB drives unplugged (so I cannot pick them by mistake)

my notes

PREPARE THE MICROSD CARD (2)

10. Steps on Windows

1. Open Imager. If Windows says "format the disk?" -> **Cancel**.
2. Device = `Raspberry Pi 4`
3. OS = `Raspberry Pi OS (64-bit)`
4. Storage = my card (check size + drive letter!)
5. Next -> edit settings (gear icon / "customise")
6. Write -> verify -> "Write successful" -> remove card



book (PDF p.36): Imager v1.8.5 on a Mac. Mine: Windows, Pi 4.

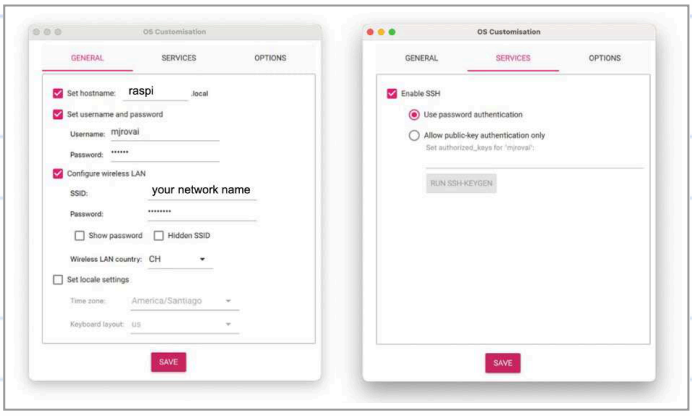
my notes

11. Settings I type (very important)

Setting	What to enter
hostname	`raspi4` (for my two Pis: `raspi-office` / `raspi-home`)
user + password	`piuser` + a strong password - write it on my connection card
Wi-Fi	name + password exactly as spelled, country `KW`
zone / keyboard	`Asia/Kuwait`, `us`
SSH	**ENABLE, password authentication** (Services tab)

Careful! Imager erases whatever drive I choose. Wrong drive = lost data.

Office Wi-Fi may need a login or block devices. Plan B: phone hotspot or an Ethernet cable.



book (PDF p.37): General tab (left), Services tab (right)

Book PDF pp. 34-37. Newer Imager versions show the same options as step-by-step screens.

my notes

FIRST BOOT & FIRST LOGIN

12. First boot (15 min)

1. Card into the slot on the **underside** (contacts toward the board).
2. Optional monitor / keyboard first.
3. Power **last** (USB-C, 5 V / 3 A). The Pi 4 has **no power button**.
4. Wait a few minutes; it may restart once. **Do not unplug!**

- RED steady = power is good
- GREEN flickers = Pi reads the card

no red light? check the power supply

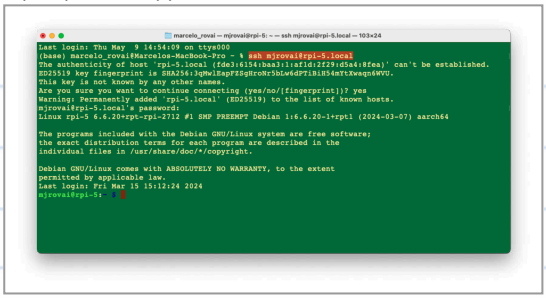
first boot: resizes the file system, applies my settings, starts SSH (1-3 min)

my notes

13. First SSH login from Windows (20 min)

- **SSH** = Secure SHell: the Pi's command line in a window on my laptop, encrypted. Same Wi-Fi needed.

1. Start -> **PowerShell**. Test: `ssh -V`
2. Connect (below). First time: type `yes`.
3. Password: **nothing shows** as I type - normal!
4. Prompt `piuser@raspi4:~ $` = I am **inside the Pi**.



book (PDF p.38): first login on a Mac

```
PS C:\> ssh piuser@raspi4.local
Are you sure you want to continue connecting (yes/no)? yes
password: (invisible)
piuser@raspi4:~ $ _
```

Book PDF pp. 38-39. Slides: Class 3, slide 7 (on Windows use cmd or PowerShell).

my notes

COMMANDS & TROUBLESHOOTING

14. My first five commands

Command	What it does / what I should see
`hostname -I`	the Pi's IP address - write it down
`uname -m`	want **aarch64** (= 64-bit); `armv7l` means 32-bit
`cat /etc/os-release`	OS version - note it
`python3 --version`	needed on Day 4 (TFLite is picky)
`free -h`	memory: close to 4 GB

ALWAYS shut down properly: `sudo shutdown -h now`
Wait until the **green LED is dark**, then unplug. Pulling the plug can corrupt the card.

my notes

15. When it does not work

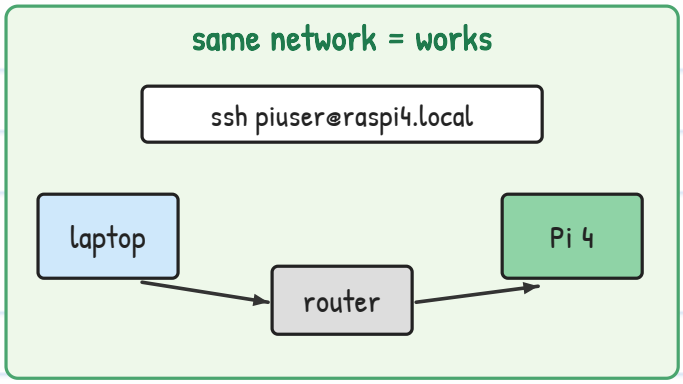
I see	probably	try
`Could not resolve hostname`	Pi not on the network yet, Wi-Fi typo, `.local` blocked	wait 5 min, check Wi-Fi, use the IP, Ethernet / hotspot
`Connection timed out`	different networks, VPN, guest Wi-Fi	same Wi-Fi, VPN off
`Permission denied`	wrong user or password	retype (case!) or re-flash
`HOST IDENTIFICATION HAS CHANGED`	I re-flashed the card	`ssh-keygen -R raspi4.local`
red LED, green never flickers	no OS found on the card	reseat, re-flash, other card
no LEDs at all	no power	other cable / charger (5 V / 3 A)

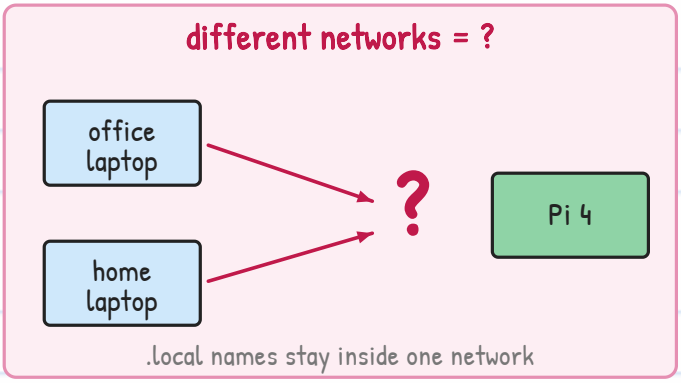
my notes

TWO PIS, TWO LAPTOPS

16. How I think about it

- Office laptop + home laptop = two **windows** into a Pi. Code and notes live on the Pi and in my GitHub repo.
- `raspi4.local` only works inside one network.





- **Pass 1:** learn on one Pi (about 2 hours). **Pass 2:** repeat on the other Pi from my notes only.
- Different host names (`raspi-office`, `raspi-home`), same user, `git pull` everywhere.

my notes

17. Quick quiz (answer here first!)

1. Image classification: fixed-function or generative?
2. Why enable SSH in Imager *before* writing the card?
3. Red LED steady = ? Green LED flickering = ?
4. What can go wrong if I unplug without shutting down?
5. Why does `raspi4.local` fail from home if the Pi is at the office?
6. Why does nothing appear while I type the SSH password?
7. Which command shows the IP? Which shows 64-bit?
8. Name three advantages of edge AI.

my notes

Answers: in the standard tutorial, "Check yourself".

MY CONNECTION CARD & WRAP-UP

18. Connection card

hostname	
user name	
password hint (not the password!)	
Wi-Fi 1 (home?)	
Wi-Fi 2 (office?)	
IP address (it can change)	
OS release	
Python version	
uname -m	
microSD size / brand	
date flashed	

my notes

19. Day 1 is done when...

- [ ] the Pi boots: red LED steady, green flickers
- [ ] PowerShell shows `piuser@raspi4:~ $`
- [ ] all five commands ran and I wrote down OS + Python
- [ ] I shut down with `sudo shutdown -h now`
- [ ] the connection card is filled in

my notes

20. Teaching it later (my ideas)

45-minute lab: 10 min big picture, 15 flash, 10 boot + LEDs, 10 SSH + commands. Typical student mistakes: wrong drive chosen, Wi-Fi typo, SSH not enabled, guest / enterprise Wi-Fi, unplugging without shutdown, "host identification changed" after re-flashing.

my notes

Tomorrow, Day 2: remote access like a pro (files, remote desktop, two networks).